

Ecology & Ecosystems

No living thing survives alone. Every organism depends on others and on its surroundings — for food, shelter, and energy. In this chapter we'll explore how living things interact, how energy and matter move through nature, and the great biomes that cover our planet.

 Georgia Standard S7L4

WHAT YOU'LL LEARN

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|---------------------------------|-----------------------------------|
| 01 Parts of an Ecosystem | 05 Cycling of Matter |
| 02 Roles Organisms Play | 06 Limiting Factors |
| 03 Feeding Relationships | 07 Earth's Biomes |
| 04 Energy Flow | 08 Putting It All Together |

4.1 Parts of an Ecosystem

An **ecosystem** is all the living and non-living things in an area, plus the way they interact. A pond, a forest, and even a rotting log are ecosystems. Every ecosystem has two kinds of parts.

Biotic Factors (living)

The living parts: plants, animals, fungi, and bacteria. Anything that is alive or was once alive.

Abiotic Factors (non-living)

The non-living parts: sunlight, water, air, soil, temperature, and rocks.

Living things constantly interact with the abiotic parts of their environment. A fish needs the right water temperature; a cactus needs sunlight and sandy soil. When the abiotic factors change, the living things must adapt — or move, or struggle to survive.

! Did You Know?

Scientists organize living things into levels too: an **organism** is one individual, a **population** is all the same species in an area, a **community** is all the populations together, and an **ecosystem** adds in the non-living parts.

4.2 Roles Organisms Play

Every organism has a job in its ecosystem based on how it gets energy. There are three main roles.

Role	How It Gets Energy	Examples
Producers	Make their own food using sunlight (photosynthesis)	Plants, algae
Consumers	Get energy by eating other organisms	Rabbits, foxes, humans
Decomposers	Break down dead things and return nutrients to the soil	Fungi, bacteria

Types of Consumers

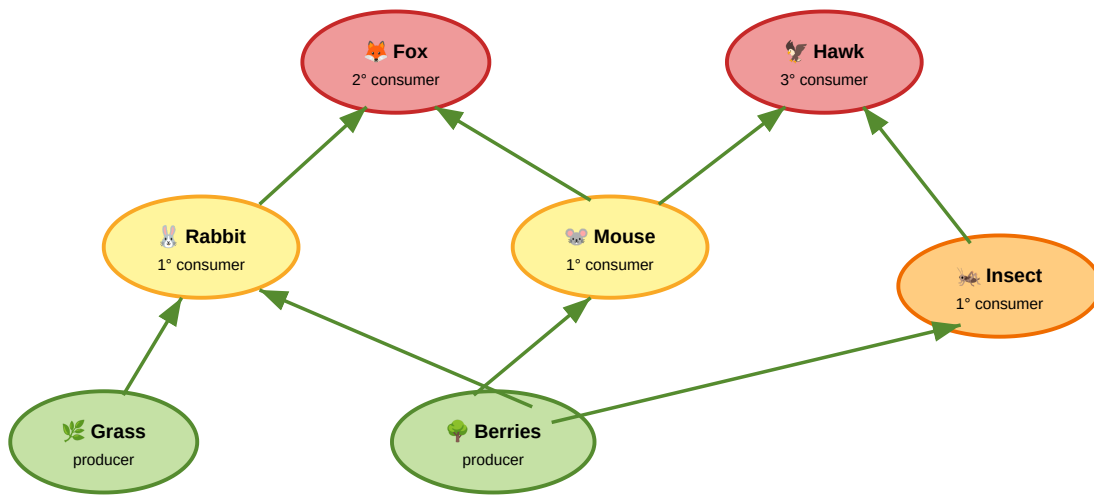
- **Herbivores** eat only plants (deer, cows).
- **Carnivores** eat only other animals (lions, hawks).
- **Omnivores** eat both plants and animals (bears, humans).

Why Decomposers Matter

Without decomposers, dead plants and animals would pile up forever and nutrients would be locked away. Decomposers recycle those nutrients so producers can use them again.

4.3 Feeding Relationships

A **food chain** shows one path of energy from producer to consumer. But in real life, organisms eat many different things — so we use a **food web** to show all the connections.



Arrows point from food → to the eater (the direction energy flows)

Figure 4.1 — A food web. Each arrow points from the food toward the organism that eats it.

★ Read the Arrows

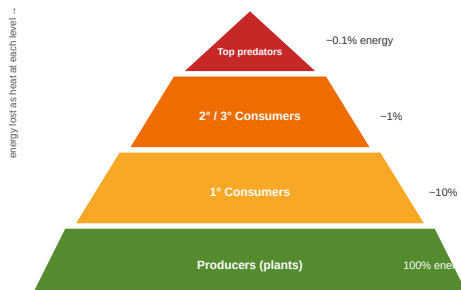
Arrows always point in the direction **energy flows** — from the organism being eaten to the one doing the eating. Grass → Rabbit means the rabbit gets energy from the grass.

✗ Common Mix-Up

Don't draw the arrow from the predator to the prey. The arrow shows where the energy **goes**, not who's hunting whom.

4.4 Energy Flow

Energy enters almost every ecosystem as sunlight. Producers capture it, and it passes up the food chain. But there's a catch: most of the energy is lost at each step.



An **energy pyramid** shows how energy decreases as you move up. Only about **10%** of the energy at one level passes to the next. The rest is used for life activities or lost as heat.

This is why there are always far more producers than top predators. It takes a huge base of plants to support just a few hawks at the top.

★ The 10% Rule

At each step up the pyramid, only about 10% of the energy moves on. If grass has 10,000 units of energy, a rabbit gets ~1,000, a fox gets ~100, and a hawk gets just ~10.

4.5 Cycling of Matter

Energy **flows** through an ecosystem and is lost, but matter **cycles** — it gets used over and over. The same carbon and water atoms have been recycled for billions of years.

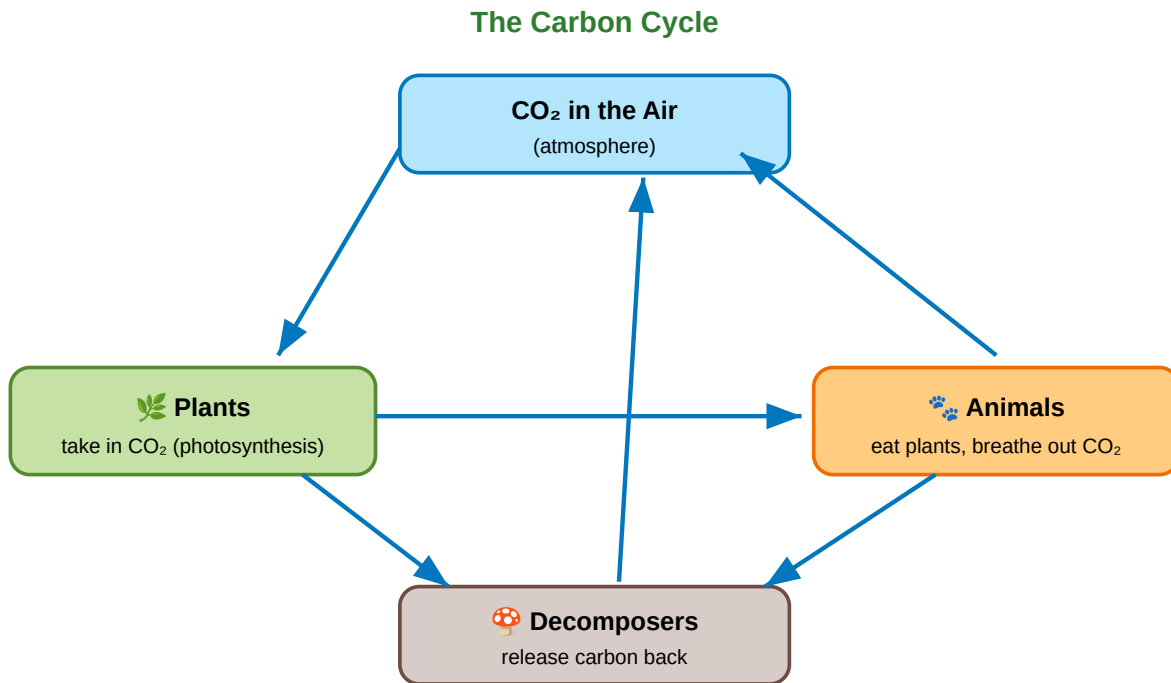


Figure 4.2 — The carbon cycle. Carbon moves between air, living things, and the soil.

Cycle	How It Moves
Water cycle	Water evaporates, forms clouds, falls as rain, and flows back — over and over.
Carbon cycle	Plants take in CO ₂ , animals eat plants and breathe out CO ₂ , decomposers return carbon to the soil and air.

🔑 Energy vs. Matter

Energy flows one direction (sun → producers → consumers) and is lost as heat. **Matter cycles** round and round and is never used up. Both are needed to keep an ecosystem alive.

4.6 Limiting Factors

No population can grow forever. A **limiting factor** is anything that keeps a population from getting too large. These factors keep ecosystems in balance.

Limiting Factor	Effect on Populations
Resource availability	Not enough food, water, or space means fewer organisms can survive.
Disease	An illness can spread quickly and reduce a population fast.
Climate	Droughts, floods, or extreme temperatures can shrink populations.
Human activity	Pollution, building, and overhunting can harm — or sometimes help — populations.

! Did You Know?

When wolves were reintroduced to Yellowstone in 1995, they controlled the elk population. That let trees grow back, which brought back beavers and birds — and even changed the path of rivers. One species can reshape an entire ecosystem.

4.7 Earth's Biomes

A **biome** is a large region with its own climate, plants, and animals. There are seven major land biomes and several aquatic ones.

 Tropical Rainforest Hot & wet all year	 Savanna Warm, grassy, dry seasons	 Temperate Forest 4 seasons, leafy trees	 Desert Very dry, hot or cold
 Grassland Grasses, few trees	 Taiga Cold, evergreen forest	 Tundra Frozen, treeless	 Aquatic Freshwater & marine

Aquatic Ecosystems

Type	Description
Freshwater	Low salt — lakes, ponds, rivers, and streams.
Estuaries	Where freshwater rivers meet the salty ocean; very rich in life.
Marine	Salt water — oceans and coral reefs, the largest ecosystems on Earth.

4.8 Putting It All Together

Ecosystems are webs of living and non-living parts. Producers, consumers, and decomposers pass energy through food webs, energy shrinks as it moves up the pyramid, and matter cycles endlessly. Limiting factors keep populations in balance, and biomes organize all of this across the planet.

Big Idea	What to Remember
Ecosystem parts	Biotic (living) + abiotic (non-living) factors interacting.
Roles	Producers make food, consumers eat, decomposers recycle.
Energy flow	Flows one way; only ~10% passes to each higher level.
Matter cycling	Water and carbon cycle and are reused forever.
Limiting factors	Resources, disease, climate, and human activity control populations.
Biomes	7 land biomes + aquatic ecosystems, each with its own climate and life.

★ Standard Check — S7L4

You can now **(a)** explain interaction patterns in ecosystems, **(b)** model energy flow and matter cycling, **(c)** analyze how factors affect populations, and **(d)** tell apart Earth's major biomes and aquatic ecosystems. Whole standard — done!